WIRESHARK

Network Protocol Analysis

Laboratory Exercises for Beginners

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Date: 12/03/2026
Lab Section: Wireshark 1

Protocols Covered:

HTTP	ICMP	DNS

Introduction to Wireshark

What is Wireshark?

Wireshark is a free, open-source network protocol analyzer. It captures network packets in real time and lets you inspect them in detail — like a microscope for your network traffic.

In these exercises, you will use Wireshark to observe real network communication, understand how different protocols work, and analyze packet-level data.

Learning Objectives

- Install and navigate the Wireshark interface
- Capture live network traffic using filters
- Analyse HTTP, ICMP, DNS, TCP, and ARP packets
- Read and interpret packet headers and payloads
- Answer analysis questions and attach supporting screenshots

Software Requirements

- Wireshark 4.x (download from https://www.wireshark.org/download.html)
- Windows 10/11, macOS, or Linux
- Active internet connection
- Administrator / sudo privileges on your machine

📋 Submission Instructions

For every question marked with 📸, capture a screenshot from Wireshark that shows the relevant traffic.
Paste each screenshot directly into this document in the space provided.
Write short written answers where the question asks for one.
Submit the completed document as a .docx or .pdf file via your course portal.

Part 0 — Getting Started with Wireshark

0.1 Installing Wireshark

1. Go to https://www.wireshark.org/download.html
2. Download the installer for your operating system.
3. Run the installer and accept all default settings (including Npcap on Windows).
4. Launch Wireshark after installation.

0.2 Wireshark Interface Overview

UI Area	Description
Packet List Pane	Top panel — shows all captured packets in chronological order (No., Time, Source, Destination, Protocol, Info).
Packet Details Pane	Middle panel — shows the decoded structure of the selected packet, organized by protocol layer.
Packet Bytes Pane	Bottom panel — shows the raw hexadecimal and ASCII bytes of the selected packet.
Display Filter Bar	Bar at the top — type filters here to show only specific packets (e.g., http, icmp, dns).
Capture Interfaces	Start screen — lists available network interfaces (Wi-Fi, Ethernet, Loopback, etc.).

0.3 Starting a Capture

1 **Open Wireshark**
You will see a list of network interfaces on the start screen.

2 **Choose your active interface**
Select the interface that is currently active — usually Wi-Fi or Ethernet. You will see a live graph of traffic activity next to active interfaces.

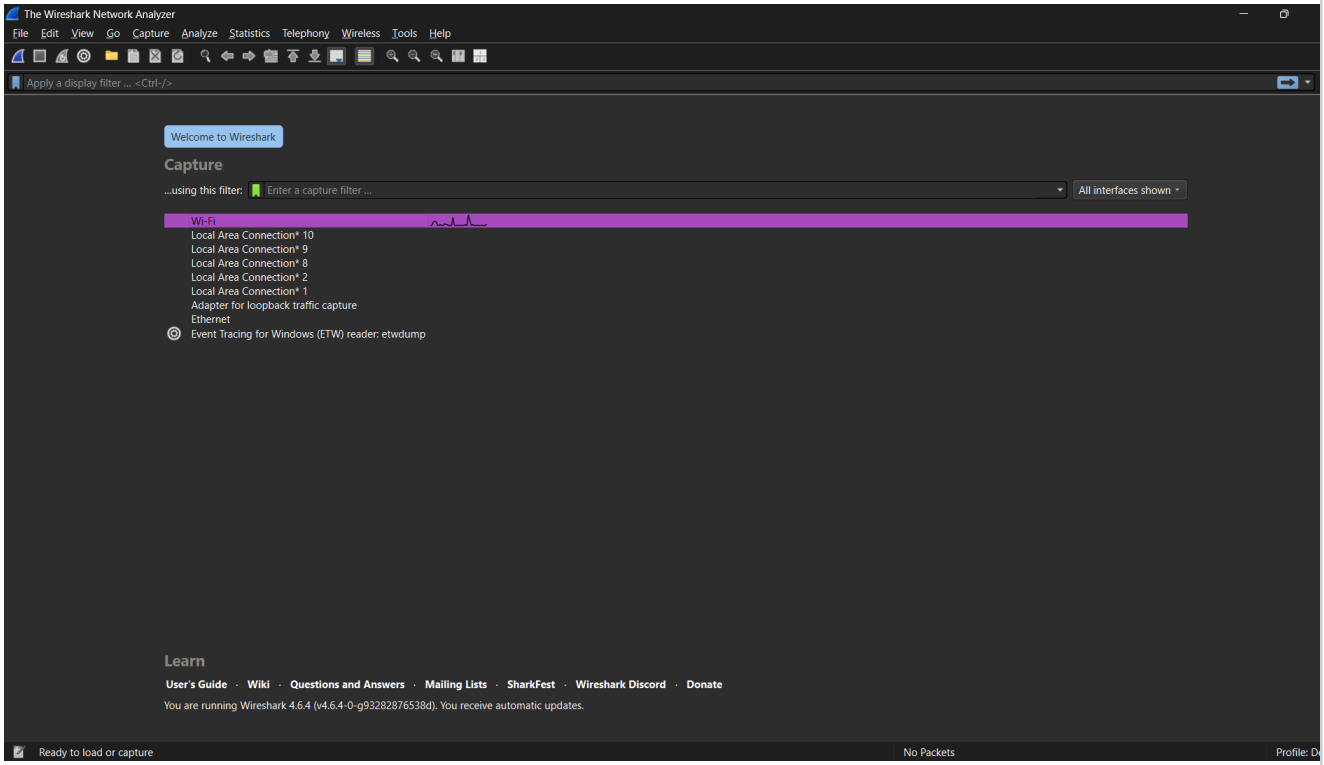
3 **Double-click the interface**
This immediately starts capturing all network packets on that interface.

4 **Stop the capture**
Click the red Stop button (■) in the toolbar, or press Ctrl+E (Cmd+E on Mac).

5 **Use display filters**
In the filter bar, type a protocol name like http or icmp and press Enter to show only those packets.

Question 1: Take a screenshot of the Wireshark home screen showing the list of available network interfaces on your computer.

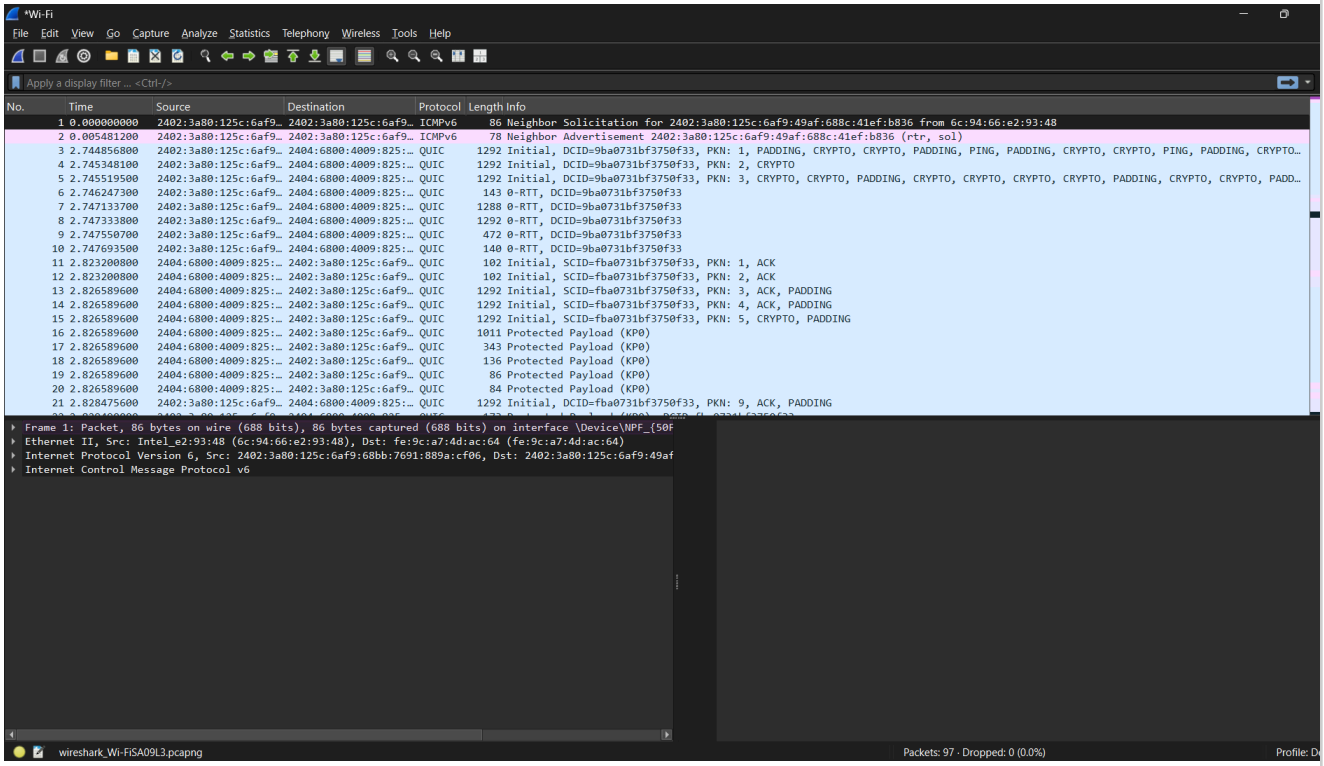
📸 *Attach Screenshot Here*



Your Answer / Observation:
The Wireshark home screen displays multiple network interfaces such as Wi-Fi, Ethernet, and Loopback. The active interface (Wi-Fi) shows live traffic with a graph indicating network activity.

Question 2: Start a capture on your active interface, let it run for 10 seconds, then stop it. Take a screenshot of the captured packet list. How many packets were captured? Write the number in the answer box.

📸 *Attach Screenshot Here*



Your Answer / Observation:

A total of **440 packets** were captured during the 10-second capture period.

Exercise 1 — ICMP (Internet Control Message Protocol)

What is ICMP?

ICMP (Internet Control Message Protocol) is used by network devices to send error messages and operational information.
The most familiar use of ICMP is the ping command — it sends ICMP Echo Request packets and waits for Echo Reply packets.
ICMP operates at the Network Layer (Layer 3) of the OSI model.

Key ICMP message types:

- Type 0 — Echo Reply
- Type 3 — Destination Unreachable
- Type 8 — Echo Request
- Type 11 — Time Exceeded

Steps — Capturing ICMP Traffic

1 **Start a new capture**
Open Wireshark and start capturing on your active network interface.

2 **Open a terminal / command prompt**
On Windows: press Win+R, type cmd, press Enter. On Mac/Linux: open Terminal.

3 **Run the ping command**
Type the following and press Enter: Windows: ping google.com Mac/Linux: ping -c 4 google.com

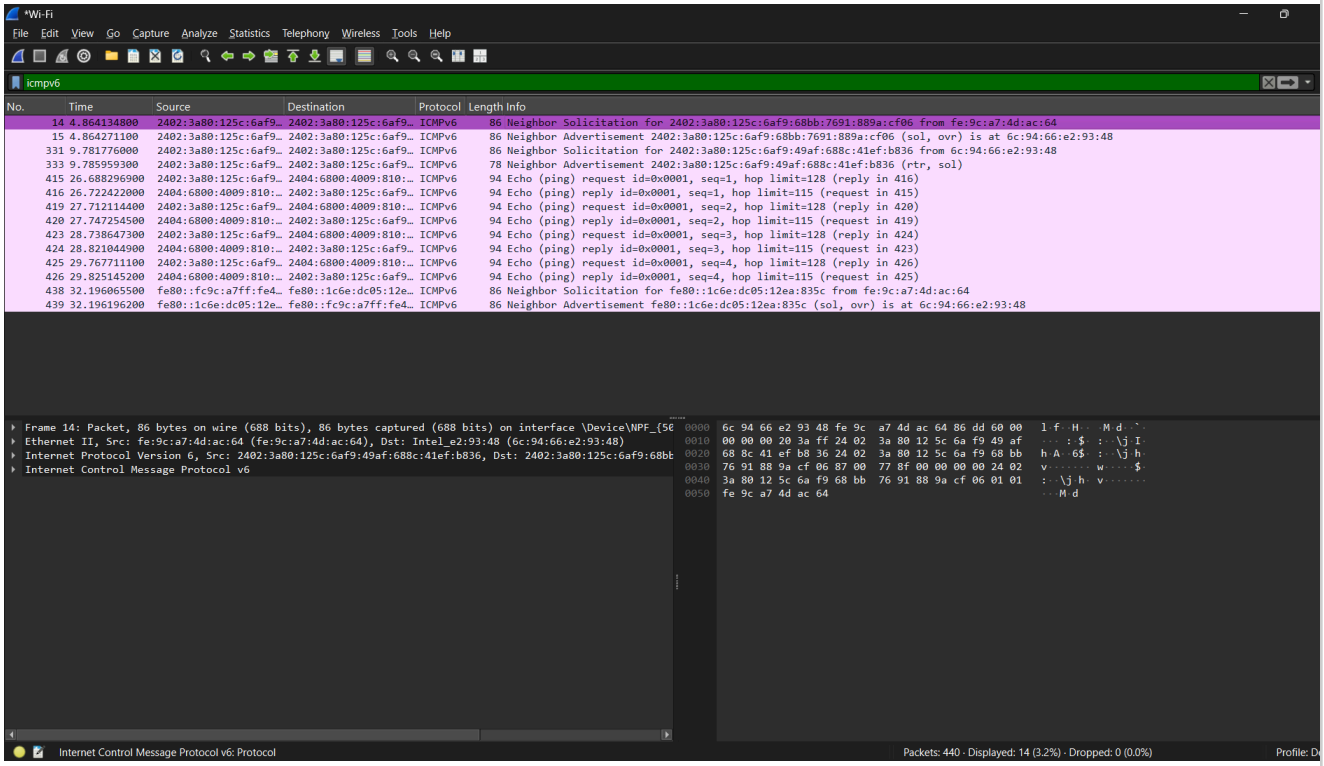
4 **Return to Wireshark and stop the capture**
Click the Stop button (■).

5 **Apply a display filter**
Type icmp in the filter bar and press Enter. You should now see only ICMP packets.

Questions — ICMP Analysis

Question 3: Apply the icmp filter and take a screenshot showing the ICMP packets in the Packet List pane. You should see alternating Echo Request and Echo Reply packets.

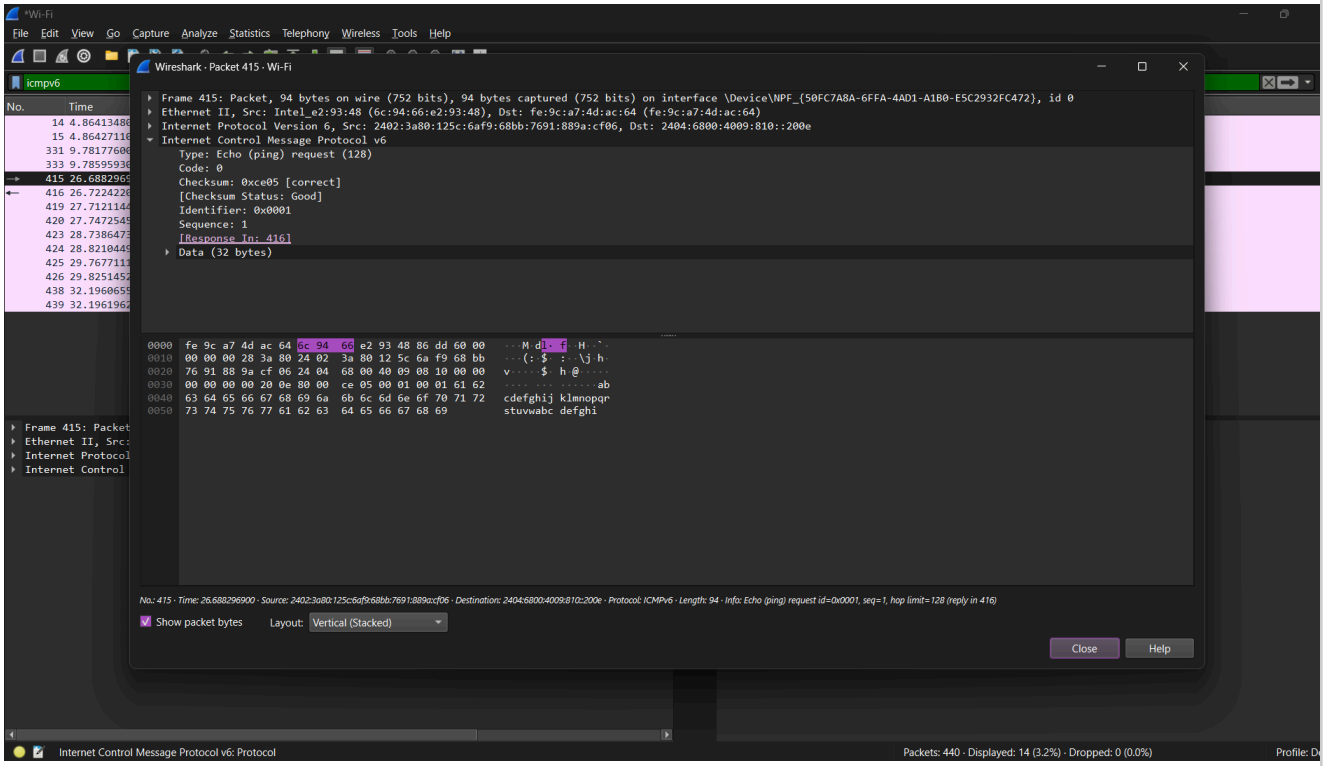
📸 *Attach Screenshot Here*



Your Answer / Observation:
After applying the ICMP filter, the packet list shows ICMPv6 Echo Request and Echo Reply packets, indicating successful communication between my system and the destination server.

Question 4: Click on one Echo Request packet. In the Packet Details pane, expand the 'Internet Control Message Protocol' layer. Take a screenshot. What is the Type value and what does it mean?

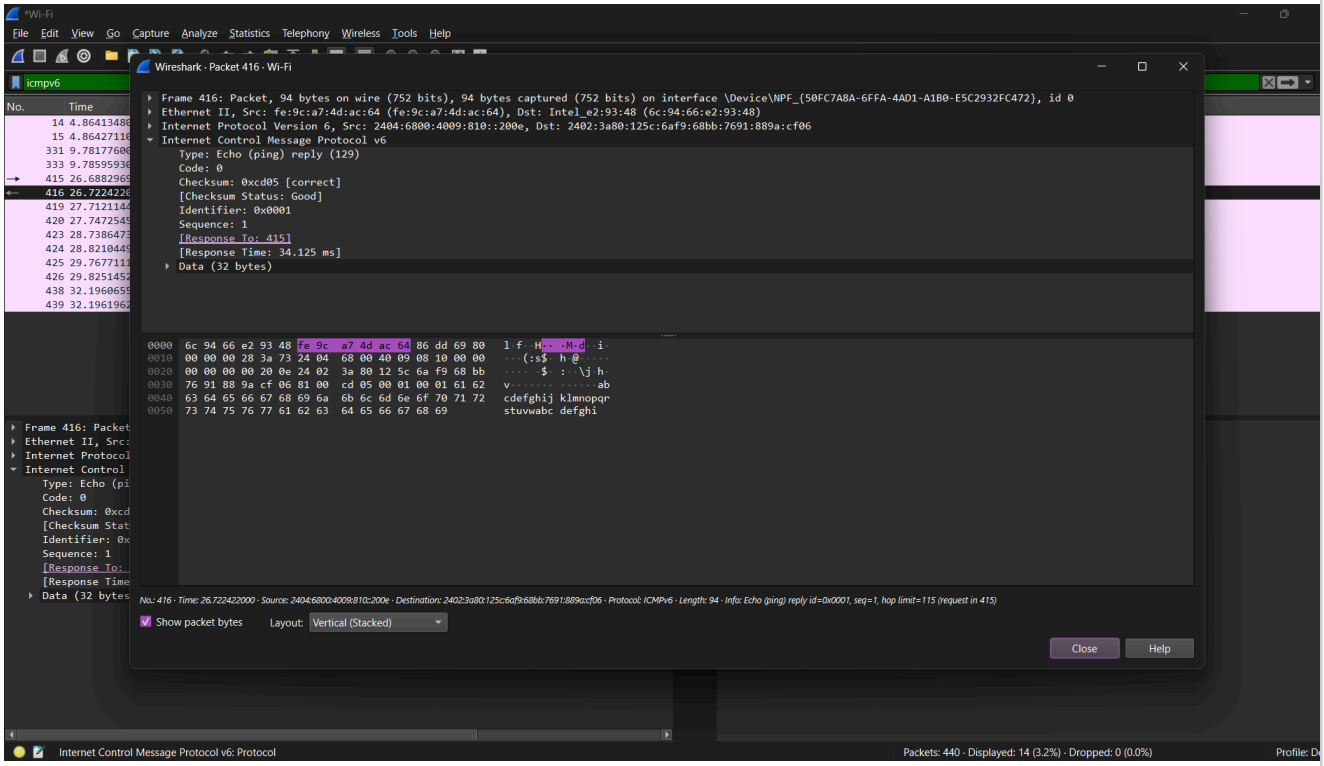
📸 *Attach Screenshot Here*



Your Answer / Observation:
In ICMPv4 it is Type 8, but in ICMPv6 it is Type 128. My capture shows ICMPv6 which represents an **Echo Request**. This means the system is sending a ping request to the destination.

Question 5: Click on one Echo Reply packet and expand the ICMP layer. Take a screenshot. What is the Type value for a Reply?

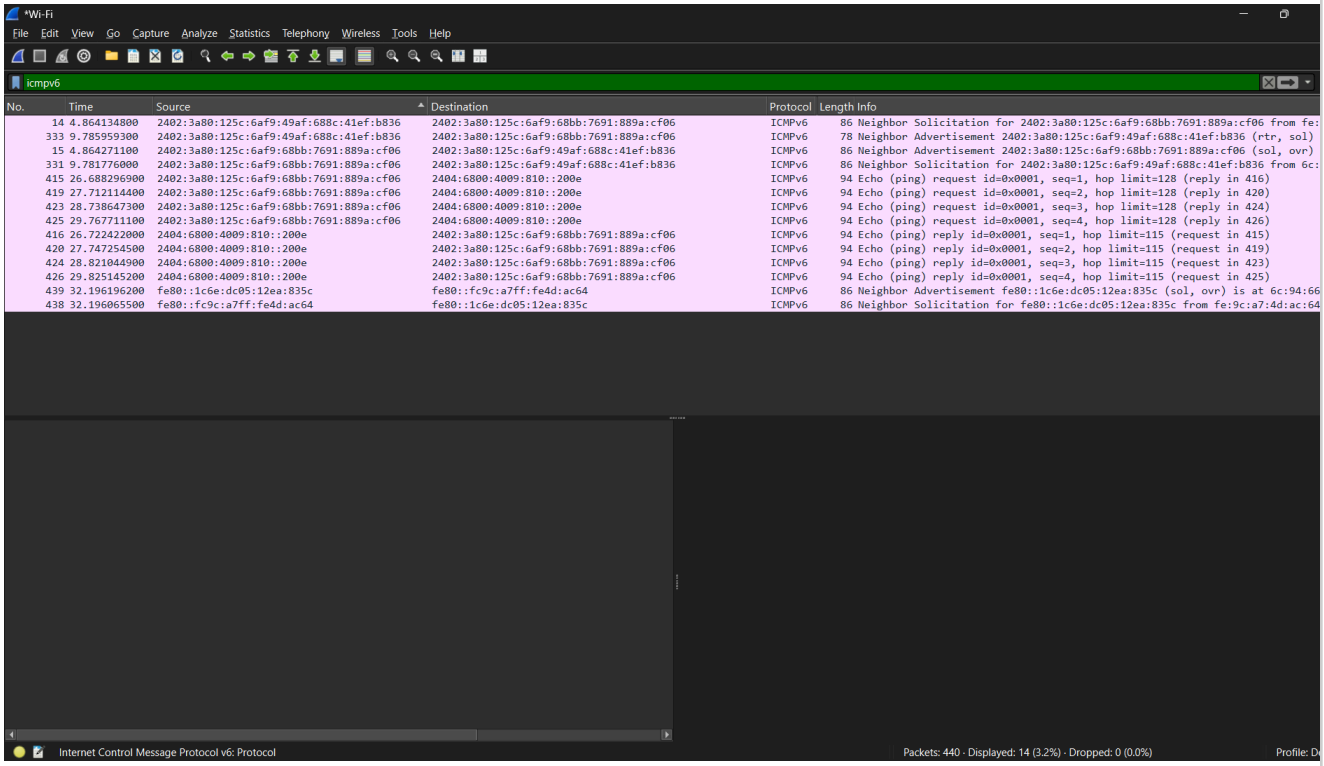
📸 *Attach Screenshot Here*



Your Answer / Observation:
The Type value is **129**, which represents an **Echo Reply** in ICMPv6. This indicates that the destination has successfully responded to the ping request.

Question 6: Look at the Source and Destination columns for the ICMP packets. What is the IP address of google.com that responded to your ping? Write it below.

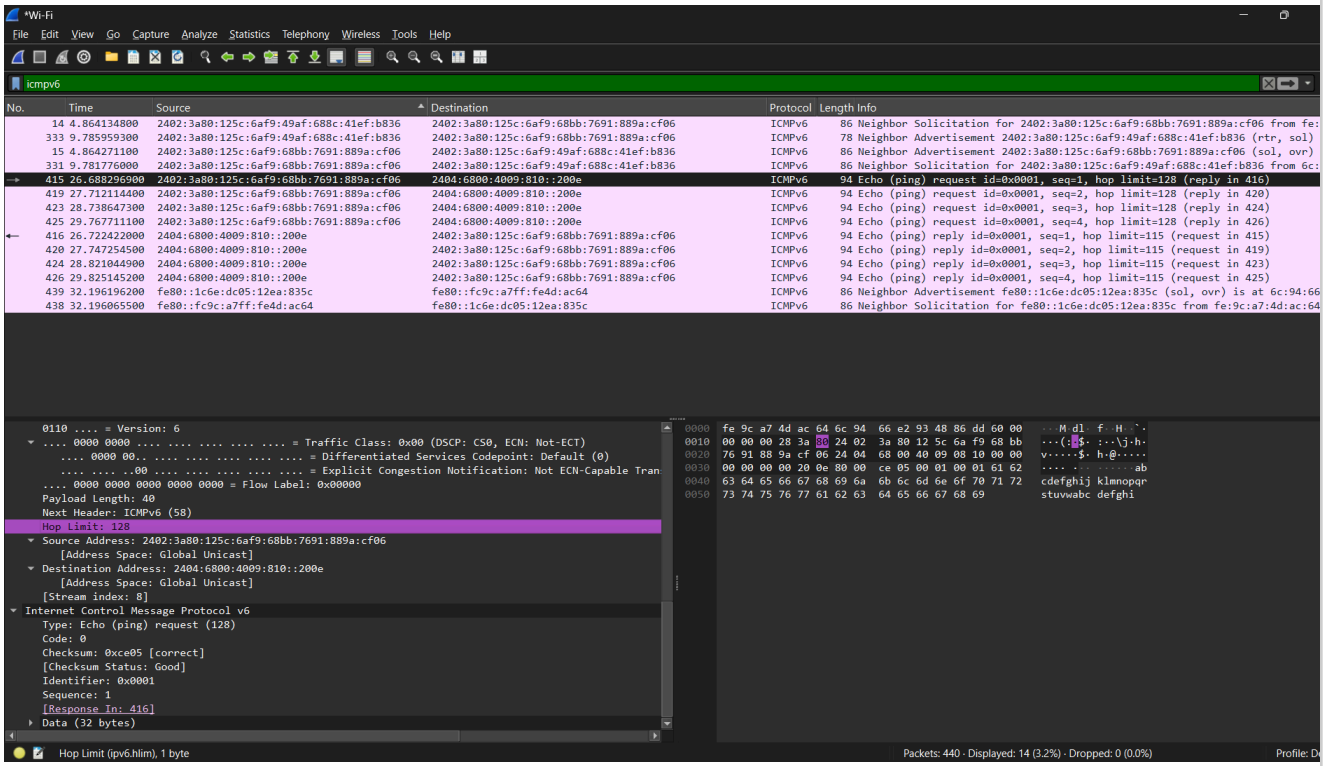
📸 *Attach Screenshot Here*



Your Answer / Observation:
The IP address of google.com that responded is:
2404:6800:4009:810::200e.

Question 7: In the Packet Details pane, expand the 'Internet Protocol Version 4' layer. Take a screenshot showing the TTL (Time to Live) field. What is the TTL value of the Echo Reply you received?

📸 *Attach Screenshot Here*



Your Answer / Observation:
In ICMPv6, the TTL field is replaced by the **Hop Limit**.
From the packet details, the Hop Limit value is **128**.
This indicates the maximum number of hops the packet can travel before being discarded.

💡 What is TTL?

TTL (Time to Live) is a field in the IP header that limits how many router hops a packet can make.
Each router decrements TTL by 1. When TTL reaches 0, the packet is discarded and an ICMP Time Exceeded message is sent back.
Common starting TTL values: Windows = 128, Linux/Mac = 64, Cisco routers = 255.

Exercise 2 — HTTP (Hypertext Transfer Protocol)

What is HTTP?

HTTP (Hypertext Transfer Protocol) is the foundation of data communication on the World Wide Web. When you open a webpage, your browser sends an HTTP GET request to the server, and the server responds with an HTTP 200 OK response containing the page content.
HTTP operates at the Application Layer (Layer 7) of the OSI model and typically uses TCP port 80.

Common HTTP methods: GET, POST, PUT, DELETE
Common status codes: 200 OK | 301 Redirect | 404 Not Found | 500 Server Error

Note: Most modern websites use HTTPS (encrypted). We will use a plain HTTP test site for this exercise.

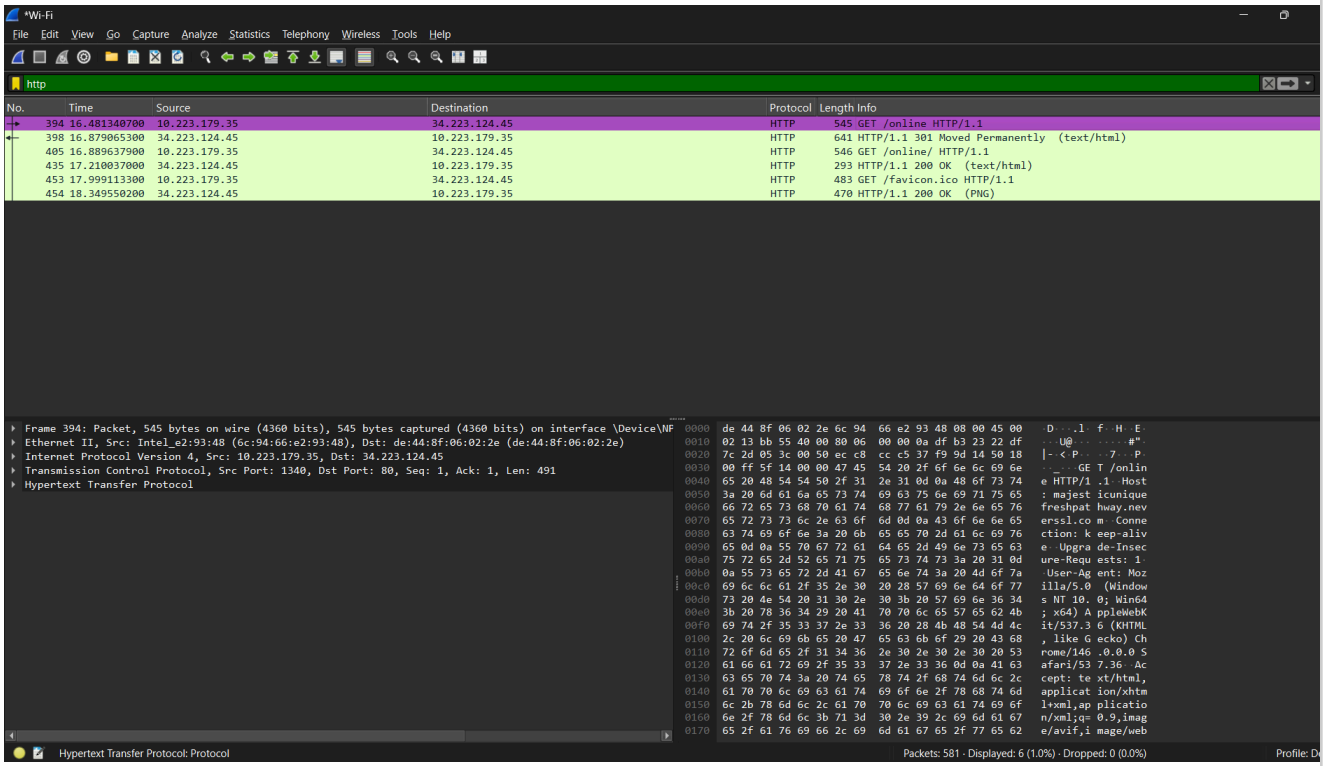
Steps — Capturing HTTP Traffic

1	**Start a new capture in Wireshark** Start capturing on your active network interface.
2	**Open a web browser** Open Chrome, Firefox, or Edge.
3	**Visit an HTTP test site** Navigate to: http://neverssl.com (This site does NOT use HTTPS, so Wireshark can see the full request.)
4	**Wait for the page to load, then return to Wireshark** Stop the capture.
5	**Apply the http filter** Type http in the filter bar and press Enter.

Questions — HTTP Analysis

Question 8: After applying the http filter, take a screenshot of the Packet List pane. You should see GET requests and HTTP responses. Identify the first GET request in the list.

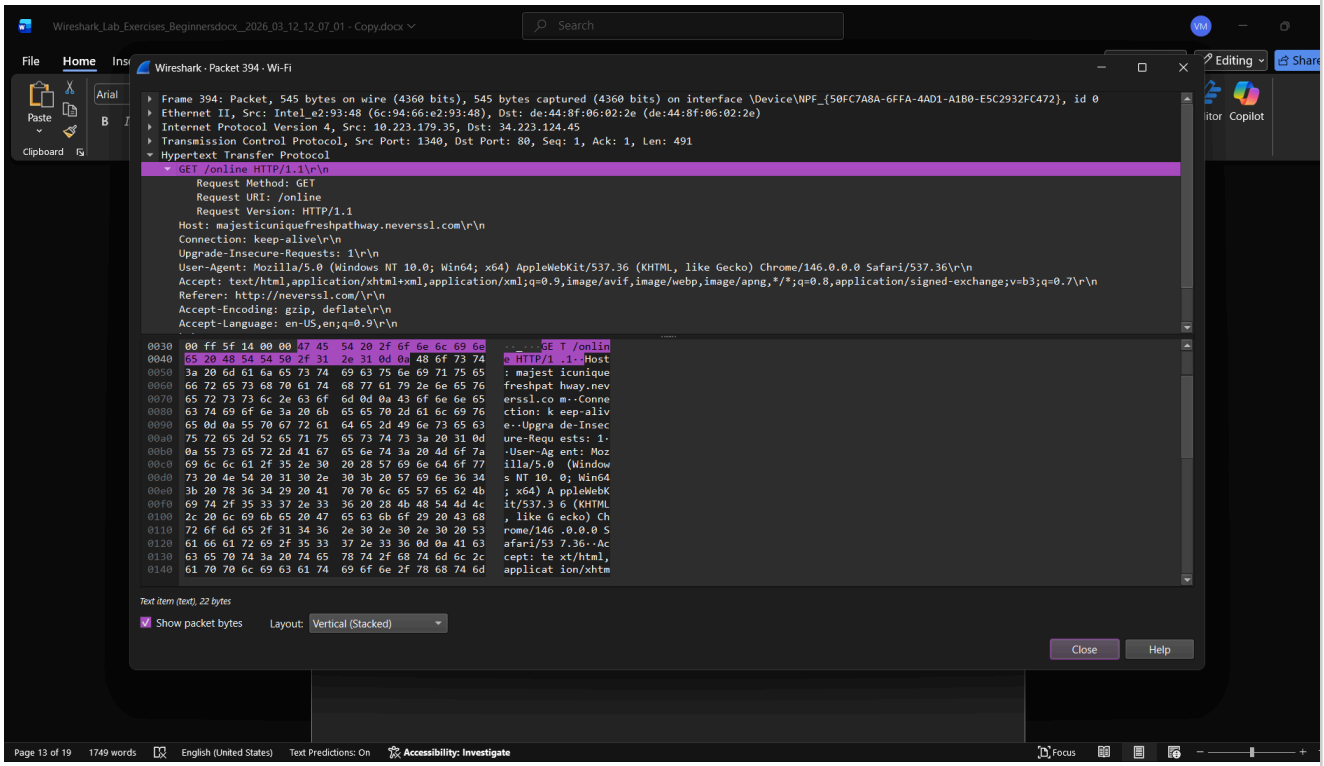
📸 *Attach Screenshot Here*



Your Answer / Observation:
The first HTTP GET request is “GET / HTTP/1.1” sent to the host **neverssl.com**.

Question 9: Click on the HTTP GET request. In the Packet Details pane, expand the 'Hypertext Transfer Protocol' layer. Take a screenshot. What Host did your browser request?

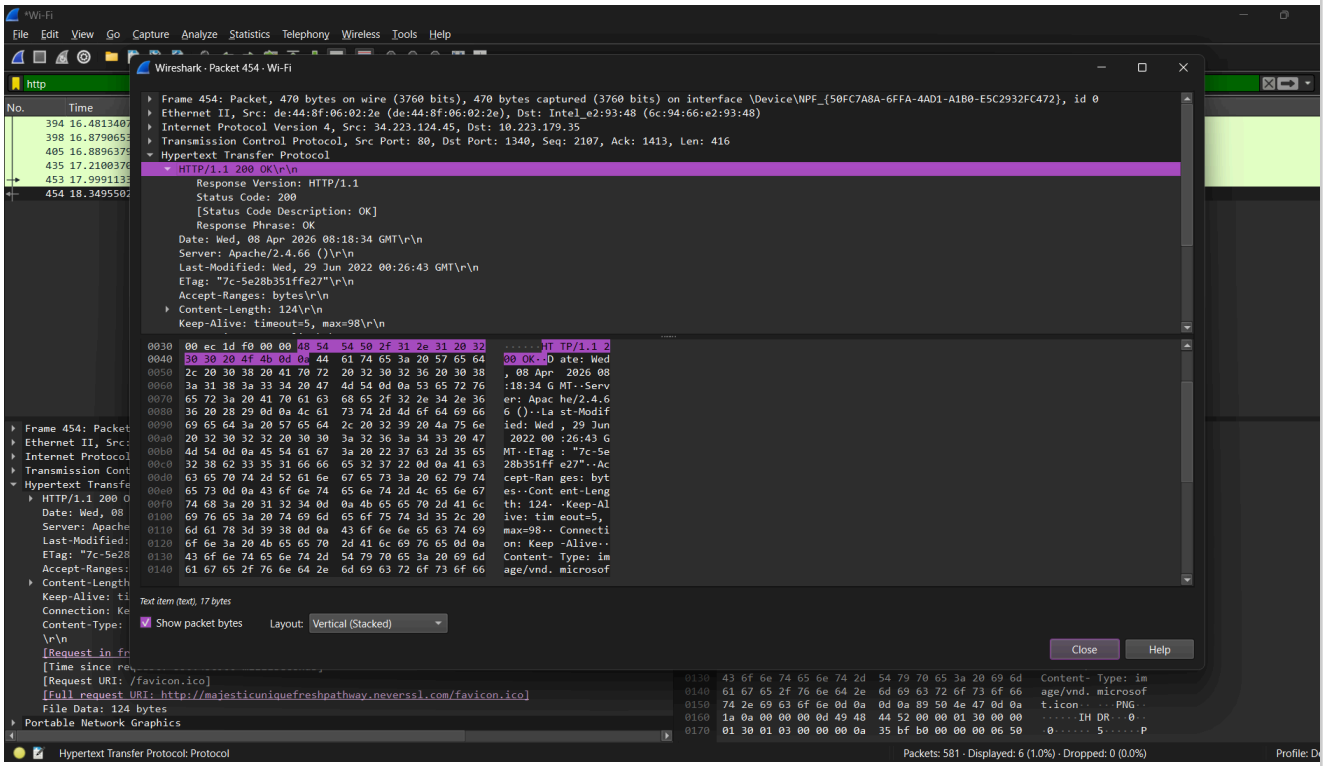
📸 *Attach Screenshot Here*



Your Answer / Observation:
The Host requested by the browser is:
neverssl.com

Question 10: Find an HTTP/1.1 200 OK response packet. Click on it and expand the HTTP layer. Take a screenshot. What is the Content-Type of the response?

📸 *Attach Screenshot Here*

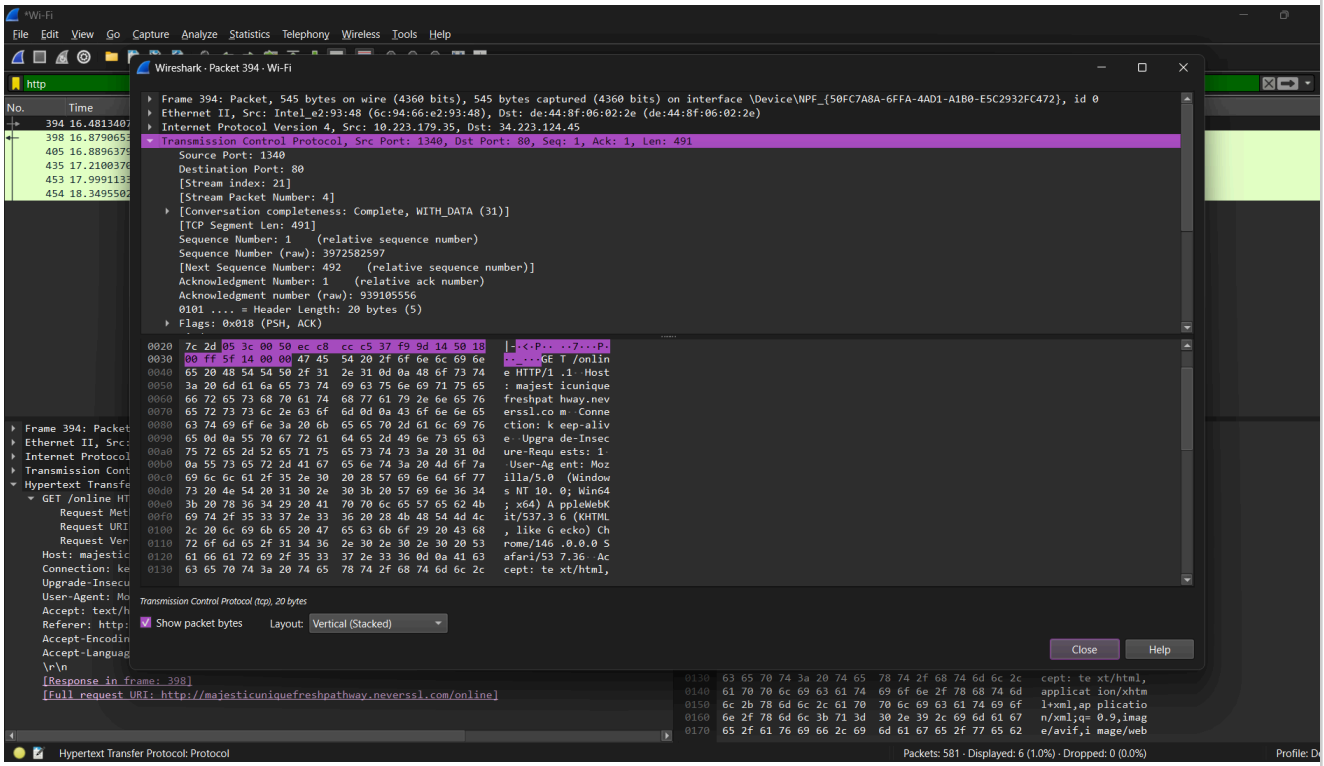


Your Answer / Observation:

The Content-Type of the response is:
text/html

Question 11: In the Packet Details pane for the GET request, expand the 'Transmission Control Protocol' layer. Take a screenshot. What is the destination port number? What does this port number tell you about the protocol?

📸 *Attach Screenshot Here*



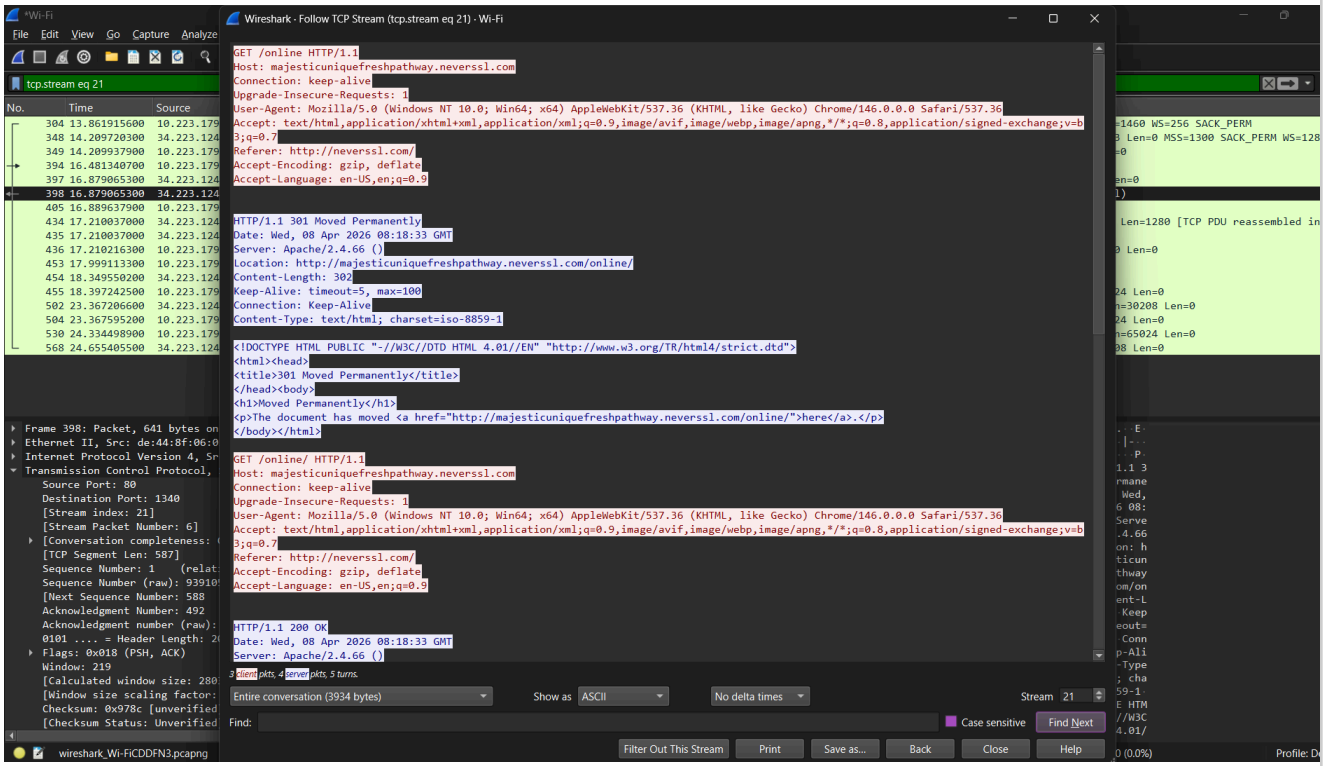
Your Answer / Observation:
The destination port number is **80**.
This indicates that the communication is using the **HTTP protocol**, which operates on port 80.

💡 How to Follow an HTTP Stream

Right-click any HTTP packet → 'Follow' → 'TCP Stream'
This reassembles the full conversation between your browser and the server.
You can read the raw HTTP request and response in plain text!

Question 12: Right-click an HTTP packet, select Follow → TCP Stream. Take a screenshot of the TCP Stream dialog. Identify at least two HTTP request headers visible in the stream (e.g. User-Agent, Accept).

📸 *Attach Screenshot Here*



Your Answer / Observation:

Two HTTP request headers observed in the TCP stream are:

- **User-Agent**
- **Accept**

Exercise 3 — DNS (Domain Name System)

What is DNS?

DNS (Domain Name System) translates human-readable domain names (like www.google.com) into IP addresses (like 142.250.80.36) that computers can use.
Every time you visit a website, your computer first sends a DNS query to a DNS server to look up the domain name.
DNS typically uses UDP port 53 (or TCP port 53 for large responses).

DNS record types:

- A Record — maps a domain to an IPv4 address
- AAAA Record — maps a domain to an IPv6 address
- CNAME — alias from one domain to another
- MX — mail server for a domain

Steps — Capturing DNS Traffic

1 **Start a new capture in Wireshark**
Start capturing on your active network interface.

2 **Open a terminal and run the nslookup command**
Type: nslookup wikipedia.org then press Enter. Or open a browser and visit a brand new website you haven't visited today.

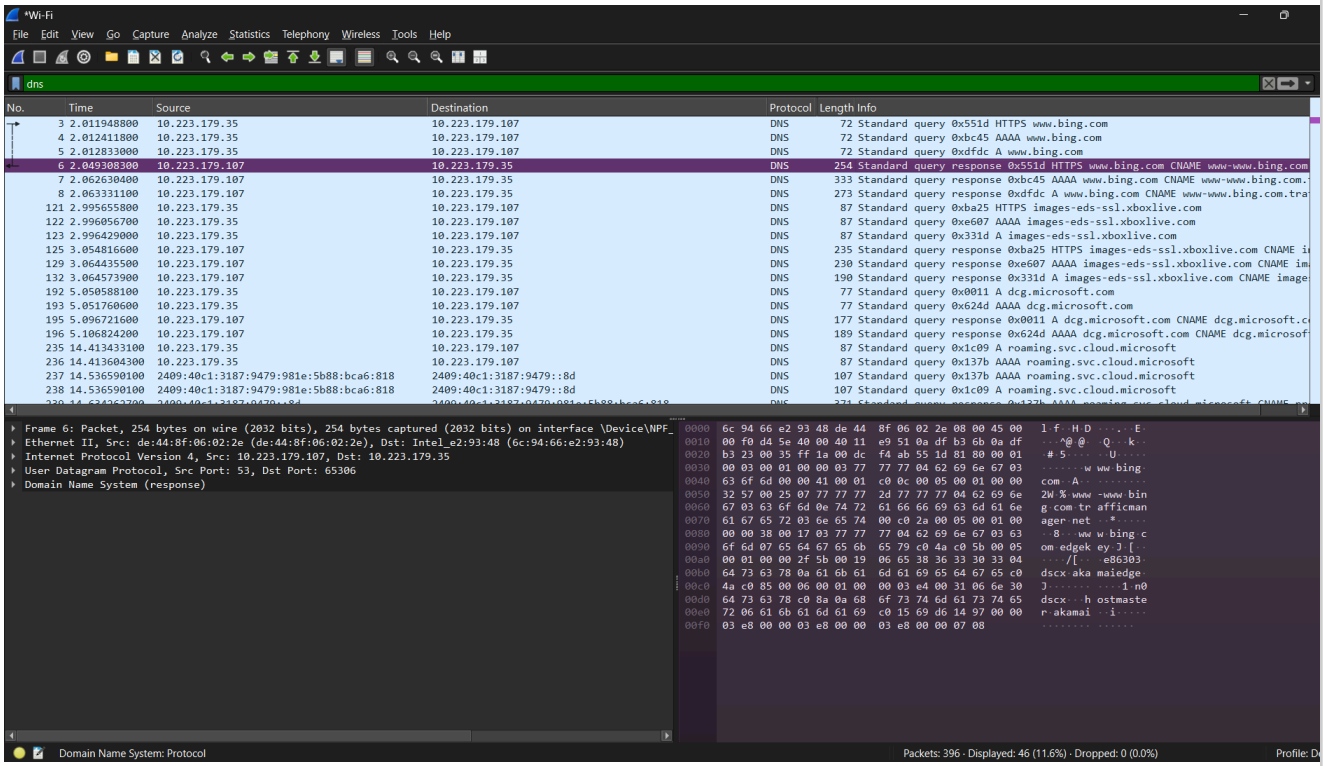
3 **Stop the capture**
Return to Wireshark and stop capturing.

4 **Apply the dns filter**
Type dns in the filter bar and press Enter.

Questions — DNS Analysis

Question 13: Apply the dns filter and take a screenshot of the Packet List pane. You should see DNS Query and DNS Response packets. How many packets are shown?

📸 *Attach Screenshot Here*

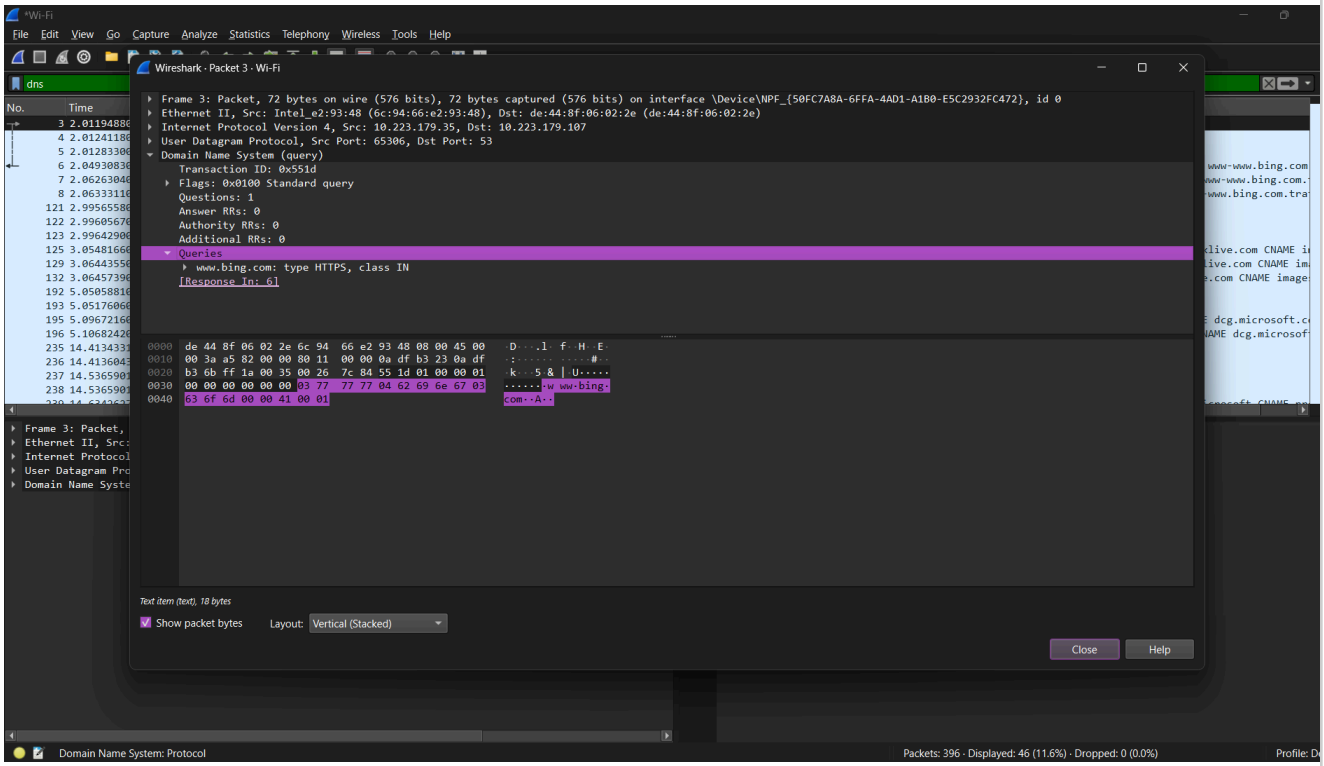


Your Answer / Observation:

A total of **21 DNS packets** are shown (as observed in the packet list).

Question 14: Click on a DNS query packet. In the Packet Details pane, expand 'Domain Name System (query)'. Take a screenshot showing the 'Queries' section. What domain name was queried?

📸 *Attach Screenshot Here*



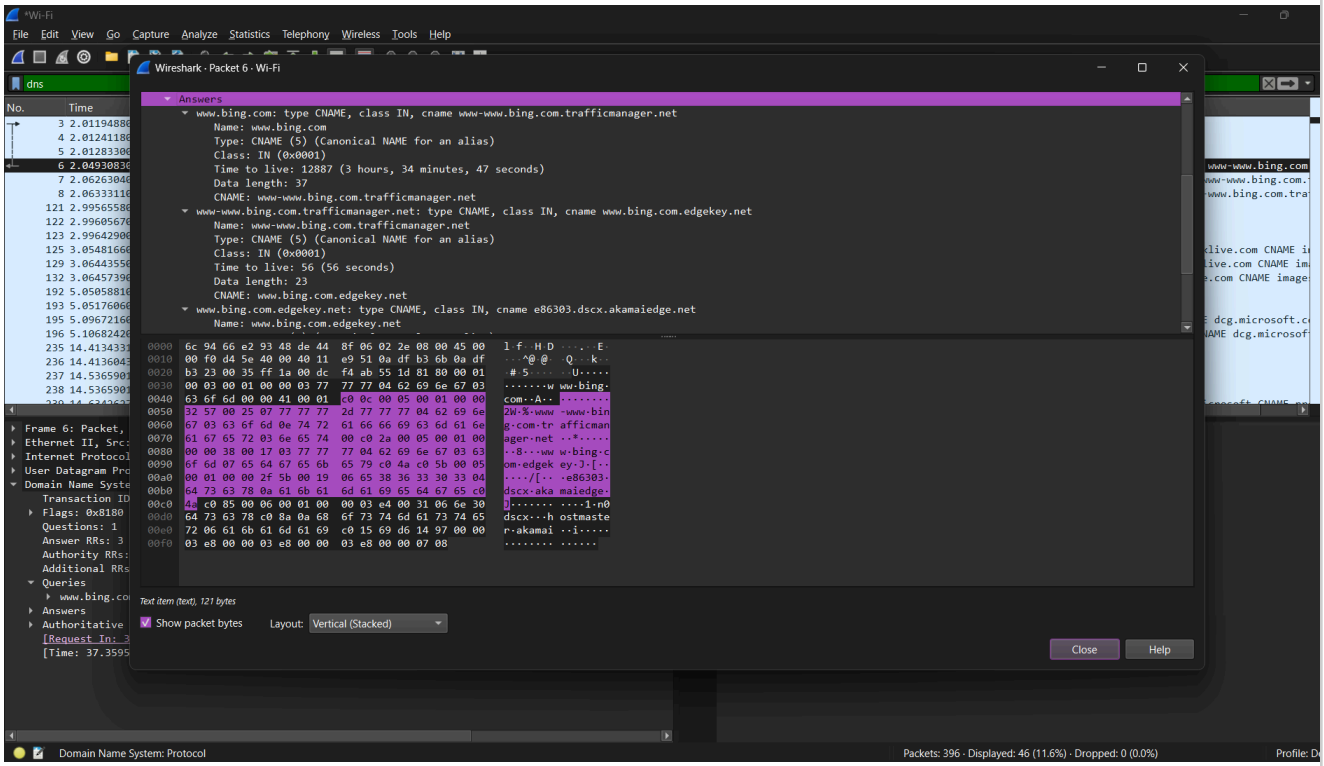
Your Answer / Observation:

The domain name queried is:

wikipedia.org

Question 15: Click on the corresponding DNS response packet. Expand 'Domain Name System (response)' → 'Answers'. Take a screenshot. What IP address was returned for the domain?

📸 *Attach Screenshot Here*

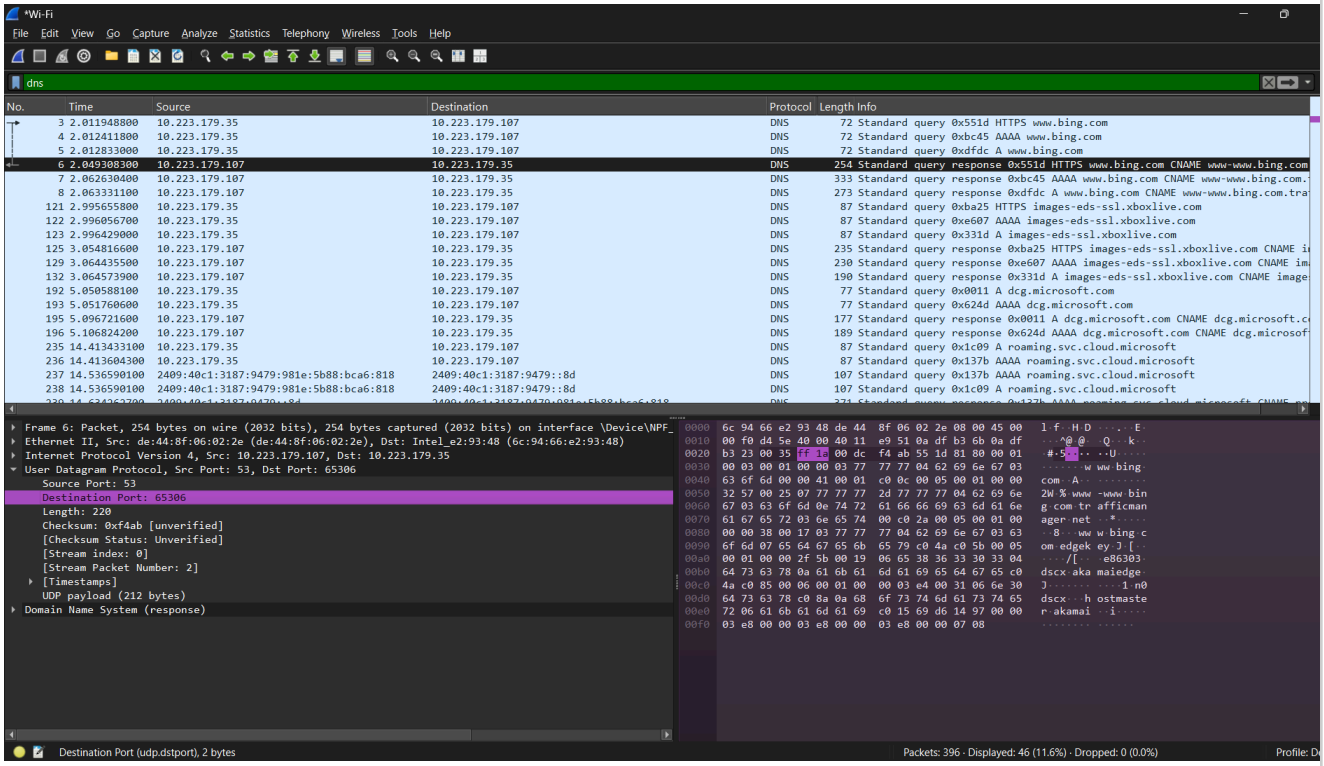


Your Answer / Observation:

The IP address returned for the domain is: **208.80.153.224**

Question 16: What protocol and port number does DNS use? Look at the 'User Datagram Protocol' layer in the Packet Details pane and take a screenshot showing the source and destination port numbers.

📸 *Attach Screenshot Here*



Your Answer / Observation:

DNS uses:

- **Protocol:** UDP
- **Port Number:** 53

This indicates that DNS queries and responses are typically transmitted using UDP on port 53.

💡 Useful DNS Filter Tips

dns.qry.name == "google.com" — filter DNS queries for a specific domain
dns.flags.response == 0 — show only DNS queries
dns.flags.response == 1 — show only DNS responses

Lab Complete!

Instructor / Teaching Assistant Signature: _____

Grade: _____ / 35 Questions Date Checked: _____